

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO5: Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.

(BTL 5)

Assessment Tool Weightage: 50%

Duration : 1 Hour

QUESTIONS: 5 Total Marks:50

Annexure A

Design Task

Code of Conduct:

I ____G Charan Teja 192372301_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Deepfake Detection and Face Authenticity Verification Platform

1. Introduction

The proposed platform is a distributed AI system designed to analyze large volumes of uploaded video content and identify deepfakes or manipulated faces. The system combines PySpark-based big-data processing, RDD operations, computer vision, metadata analysis, intelligent agents, and machine learning models. Videos are processed in parallel so that the platform can scale to thousands of files while maintaining efficient detection and verification.

2. System Architecture

The architecture contains several layers. First, a video ingestion layer receives uploaded videos and stores them in distributed storage. A PySpark processing layer divides the data into partitions and uses RDD operations such as `map()`, `filter()`, `reduceByKey()`, and `join()` to process frames and metadata in parallel. The AI analysis layer extracts faces, facial landmarks, blink patterns, lighting inconsistencies, compression artifacts, and temporal abnormalities. A metadata module checks information such as creation time, device details, codec, location data, and editing history. The results are sent to a decision engine that calculates a final authenticity score. Suspicious videos are stored in a prioritized human-review queue.

3. PySpark and RDD Operations

PySpark enables distributed processing of large video datasets. Each video can be split into frame batches and represented as RDD records. `map()` is used to extract features from frames, `filter()` removes invalid or low-quality frames, `reduceByKey()` aggregates detection scores for each video, and

join() combines visual analysis with metadata results. This approach allows multiple worker nodes to analyze different parts of the dataset simultaneously. Cached RDDs can also improve performance when extracted features are reused by several detection models.

4. Data Fusion Techniques

The platform uses multi-modal data fusion to improve accuracy. Visual features include face boundaries, skin texture, unnatural eye movement, lip synchronization errors, GAN artifacts, lighting differences, and frame-to-frame inconsistencies. Metadata features include file format, codec, timestamps, camera information, software tags, and possible signs of editing. Feature-level fusion combines normalized visual and metadata features into one vector for a machine learning classifier. Score-level fusion combines confidence scores from multiple deepfake models using weighted averaging or ensemble methods. The final score provides a stronger authenticity decision than relying on a single technique.

5. Intelligent Agent Coordination

Specialized intelligent agents coordinate the workflow. An ingestion agent validates uploaded files, while a visual analysis agent examines facial and frame artifacts. A metadata agent checks file-level information and consistency. A fusion agent combines the outputs of different detectors. A risk agent calculates a suspiciousness score based on detection confidence, source reputation, and possible manipulation severity. Finally, a queue-management agent prioritizes high-risk and high-impact videos for human review. Agents can exchange results through a shared message queue or distributed database, allowing the system to operate efficiently and independently.

6. Review Prioritization and Continuous Retraining

Suspicious content is ranked using a priority score. Videos with high deepfake confidence, multiple inconsistent signals, or large potential impact are reviewed first. Human reviewers can confirm whether content is authentic or manipulated. Their feedback becomes labeled training data for continuous model improvement. New samples are periodically processed using PySpark, and updated datasets are used to retrain and evaluate detection models. Model performance is monitored using measures such as accuracy, precision, recall, and false-positive rate. If performance decreases because of new deepfake techniques, the platform can trigger retraining and deploy improved models.

7. Conclusion

This distributed platform provides a scalable solution for deepfake detection and face authenticity verification. PySpark and RDD operations support large-scale parallel processing, while data fusion combines visual artifacts and metadata for better decisions. Intelligent agents automate analysis, risk assessment, and review prioritization. Continuous human feedback and model retraining help the platform adapt to evolving deepfake generation methods and maintain reliable detection performance.